

2 Results

In this section, we introduce the definition of the model of interest along with the relevant notation. We then proceed to state the main results of this work.

2.1 Definition of the model

Throughout, we use the notation $\mathbb{N} := \{1, 2, \dots\}$, $\mathbb{N}_0 := \mathbb{N} \cup \{0\}$ and $[n] := \{1, \dots, n\}$ for each $n \in \mathbb{N}$. We consider a labeled directed graph (V, E) with vertex set $V = [n]$ for some $n \in \mathbb{N}$ and edge set $E \subset \{(v, w) : v, w \in V\}$. The natural number representing a vertex is then referred to as its **label**. In such a graph, a sequence of vertices $(v_1, v_2, \dots, v_{k+1})$ is a **directed path** of length $k \in \mathbb{N}$ if

$$\forall i \in [k] : (v_i, v_{i+1}) \in E$$

and an **undirected path** of length $k \in \mathbb{N}$ if

$$\forall i \in [k] : (v_i, v_{i+1}) \in E \vee (v_{i+1}, v_i) \in E.$$

A labeled directed graph is called

- **connected** if, for each pair of vertices, there is an undirected path joining those vertices,
- **acyclic** if it does not contain any directed cycles, i.e. directed paths that visit the same vertex more than once,
- **increasing** if the labels are decreasing along all directed paths.

We observe that if a graph is increasing, then it is acyclic in particular. For $v \in V$, we call a vertex $w \in V$ an **out-neighbor** if $(v, w) \in E$ and an **in-neighbor** if $(w, v) \in E$. Then, the number of out-neighbors of v is denoted by **outdeg**(v). On the other hand, we call the number of in-neighbors of v the **degree** of vertex v , denoted by $d_V(v)$. For all $n, m \in \mathbb{N}$ we define $\mathcal{I}_n^{(m)}$ to be the set of increasing directed graphs on $[n]$ satisfying the condition

$$\forall v \in [n] : \text{outdeg}(v) = m \wedge (v - 1). \quad (2.1)$$

Now, we are ready to formally define RRDAGs.